

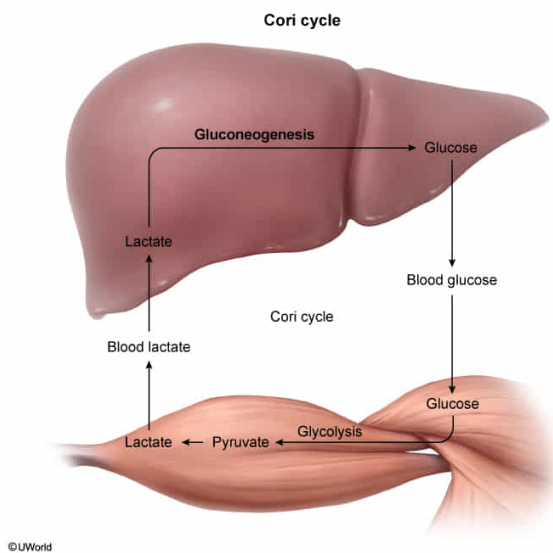
During anaerobic exercise, the Cori cycle connects which two metabolic pathways?

- ☐ A. Glycogenolysis and the citric acid cycle [12%]
- ☐ B. The urea cycle and gluconeogenesis [21%]
- ☐ C. The pentose phosphate pathway and glycogenesis [12%]
- ✓ ☒ D. Gluconeogenesis and glycolysis [52%]

Correct

53% answered correctly

### Explanation:



During **glycolysis**,  $\text{NAD}^+$  is converted to NADH by the enzyme glyceraldehyde-3-phosphate (GAP) dehydrogenase. For glycolysis to continue,  $\text{NAD}^+$  must be **regenerated**. Under aerobic conditions, electrons from NADH can be transferred to the electron transport chain (ETC) and ultimately to oxygen. However, in anaerobic conditions,  $\text{NAD}^+$  cannot be regenerated by the ETC because there is insufficient oxygen to accept electrons. Consequently, NADH donates electrons to pyruvate, which is reduced to lactate in the process. **Lactate that builds up** from this mechanism must be removed from the system because it can lead to muscle pain and nausea.

The lactate in **muscles** enters the **bloodstream**, which carries it to the **liver**. In the liver, lactate is converted to glucose during **gluconeogenesis** and is carried back to muscles by the blood. The process of carrying lactate from the muscle to the liver and moving regenerated glucose from the liver back to muscles is called **the Cori cycle**, which **connects gluconeogenesis and glycolysis**.

**(Choice A)** **Glycogenolysis** is the consumption of glycogen to form glucose-6-phosphate, and the **citric acid cycle** is the consumption of acetyl-CoA (formed from pyruvate) to generate  $\text{CO}_2$ , GTP, NADH, and  $\text{FADH}_2$ . Neither pathway is part of the Cori cycle.

(Choice B) The **urea cycle** processes nitrogen from amino acids for excretion, and gluconeogenesis generates glucose from pyruvate and other precursors. Both pathways occur primarily in the liver, and they are not connected by the Cori cycle.

(Choice C) The **pentose phosphate** pathway produces ribose 5-phosphate and NADPH using glucose as a starting material. **Glycogenesis** is the production of glycogen from glucose. The two pathways diverge from each other and are not connected by the Cori cycle.

**Educational objective:**

During anaerobic exercise, pyruvate is reduced to lactate to regenerate  $\text{NAD}^+$ . Lactate that builds up in muscles is sent to the liver, where it is converted back to glucose and returned to muscles. This process is called the Cori cycle.

**Time spent:**0 sec

**Subject:**  
Biochemistry

**QID:**402133

**System:**  
Metabolic Reactions